

Possibilistic Decomposition of a Tomographic Inversion

Separating data-forced structure from regularization artifact

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What this is

A tomographic image is not a picture of the Earth. It is the output of a long chain of lossy, selective steps—sparse rays, a simplified forward model, a parameterization, a regularizer, an optimizer—and what survives that chain is not the structure but a *filtered residue* of it. You and I have said as much to each other already. The practical question that framing forces is the one this note is about:

Given a finished inversion, which features did the data force, and which did the regularization invent?

Standard practice does not answer that question. It picks a damping value, returns one model, and attaches a posterior covariance—and every part of that is conditional on the damping choice. Your own thesis says it plainly: “there is no simple solution for regularization, and optimization of damping conditions remains a highly parametrization and input dependent problem” (ZTM/TFM, §11).

This note proposes a different reading of an inversion—a **possibilistic** one—and demonstrates it end-to-end on synthetic travel-time tomography with two forward operators, straight-ray and Eikonal. It is a methods communication and a draft; the demonstrations are synthetic; nothing here is claimed as proven. What I am claiming is that the reading is sound, that it is implemented and validated against known ground truth, and that the place it stops working cleanly is a real open problem worth our working on together.

1 The problem: a damping choice is not a fact

Tomographic inversion is ill-posed. Many models fit the data within noise. To return *one* model you must regularize—damp, smooth, prefer a reference—and the model you get is as much a property of that choice as of the data.

The honest consequence: a feature in a tomographic image belongs to one of three classes, and they are not the same kind of thing.

A posterior covariance does not draw this line. It reports a spread *under one prior and one*

Class	Meaning
Forced	Present in <i>every</i> model consistent with the data and the hard physical bounds. Data-determined. Independent of the damping choice.
Forbidden	Present in <i>no</i> such model.
Measure-dependent	Present in <i>some</i> consistent models and absent in others. The data do not determine it—which sign or structure you see is set by the regularization. Such a feature may still be physically real: measure-dependence diagnoses <i>underdetermination</i> , not falsehood.

damping; it cannot tell you that a given feature would vanish under a different, equally defensible choice. The damping problem of §11 is, in this language, a symptom of reading an inversion through the wrong layer: it is the search for the “right” measure-layer choice to extract a model, when the features that *depend* on that choice are exactly the ones the data does not determine.

2 The possibilistic frame

The distinction in §1 is the **possibilistic** / **probabilistic** split, taken from the two-layer discipline of the Closure Forces Structure programme (`inverse_born_methodology.md`) and carried over to inversion:

- The **possibilistic layer** asks what is *forced or forbidden* by the data and the hard physical constraints alone. Its answers are unconditional—they do not depend on a prior, a damping value, or a measure.
- The **probabilistic layer** asks how a measure distributes weight over what the possibilistic layer permits. Its answers are conditional on that measure.

A posterior covariance is a probabilistic-layer object. The possibilistic layer is the one that answers the §1 question, and it is the one standard tomography leaves on the table.

One clarification carries the rest. Throughout, F is the *hard* admissibility set—every model that fits within the noise tolerance and obeys the hard physical bounds—and the probabilistic layer is taken as conditional on that admissibility. With a smooth likelihood a posterior need not vanish sharply outside F ; the two-layer reading treats F as the tolerance-defined region and a measure μ as a measure on it. Every “measure on F ” below—and every bound stated in terms of “ F ’s extent,” including Figure 3’s—is a claim about admissible μ in that conditional sense, not about an untruncated smooth-likelihood posterior.

The split has a natural geometry, and the next three figures build it up one idea at a time.

Figure 1 — the feasible set. The data and the hard constraints do not pick out a model; they pick out a *set* F of models—every model that fits within noise and respects the bounds. F is a set, not a probability density—the figures shade it only to mark the region. Project F onto a feature’s axis and the feature falls into one of two classes. It is **forced** when the projection lies entirely on one side of zero: every model consistent with the data agrees on its sign, and no prior is needed to settle it. It is **measure-dependent** when the projection straddles zero: the data

leave the sign open, and any single value reported there is fixed by the measure you add, not by the data.

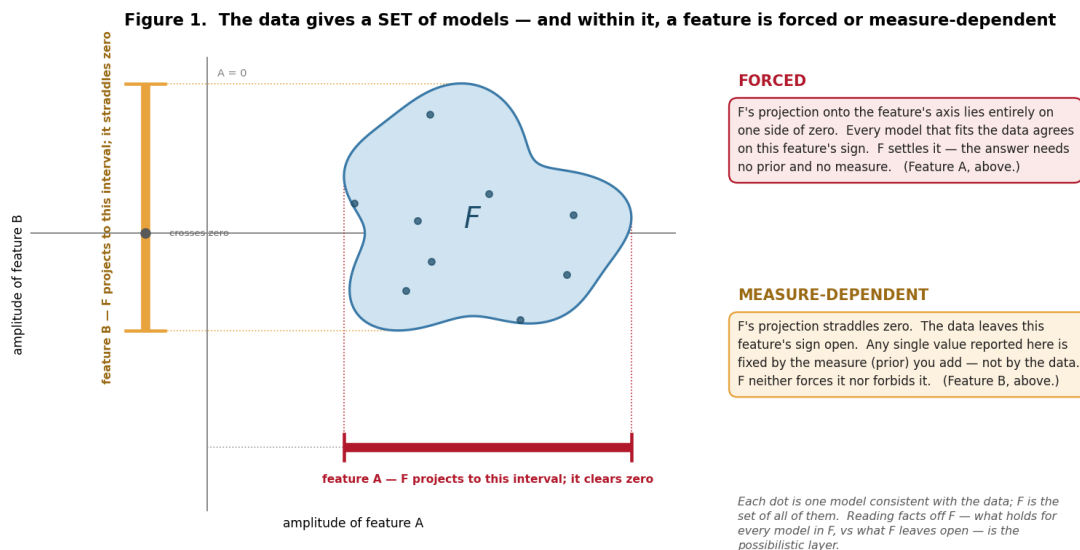


Figure 1: The data give a set, not a point. A feature is forced when F 's projection clears zero and measure-dependent when it straddles zero—the two terms this note turns on, defined geometrically.

On the word “possibilistic.” I use it in a deliberately narrow sense—the crisp-set limit of possibility theory, equivalently a set-valued feasibility reading, not a graded possibility distribution. F is an ordinary feasible set: a model is in it or it is not. A feature is *forced* exactly when a fact holds for every model in F (necessity 1, in possibility-theory terms) and *measure-dependent* when F neither forces nor forbids it. No fuzzy membership is invoked; the only structure used is the set F and the all-or-nothing quantifier over it.

Figure 2 — two reports. The same F can be turned into a reported answer two ways. The *possibilistic report* uses exactly F : forced features pinned to a sign, measure-dependent features returned as intervals—no more information than F carries, and no less. The *Bayesian report* uses F plus a measure μ —it places μ on F and reports its mean and credible interval. μ is information beyond F , and two defensible choices of μ can disagree on a measure-dependent feature's sign, so that choice needs a justification of its own. On a forced feature the two reports agree; the divergence is confined to the measure-dependent ones.

Figure 3 — the layers compose. The two are not rivals run side by side; they compose in sequence. The possibilistic layer runs first and bounds which measures are admissible—a measure is admissible only if it is supported on F . The Bayesian layer then runs inside that bound: across all admissible measures, the *attainable range of reported answers*—the measure-uncertainty—cannot exceed F 's extent. On a forced feature that bound clears zero, so no admissible measure can move the sign; on a measure-dependent feature it straddles zero and the sign is genuinely open. Possibilism does not compete with Bayes—it brackets it.

The object that carries the possibilistic content is the **feasible interval**. Take an ensemble of models that each fit the data within noise and respect the hard bounds. For each cell, record the

Figure 2. The same feasible set F — two reports

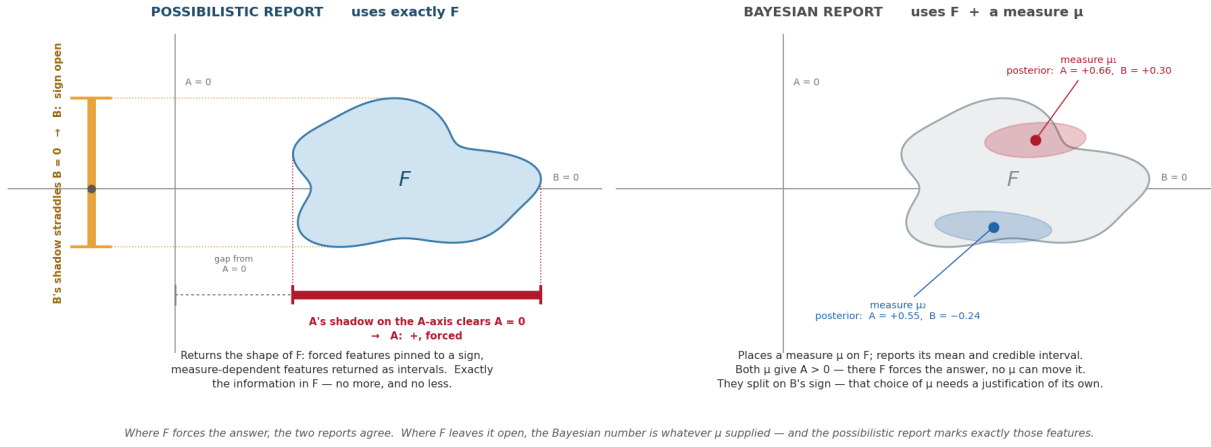


Figure 2: Two reports from one feasible set. Possibilistic: use exactly F . Bayesian: use F plus a measure μ —defensible μ agree where F forces the answer and can conflict where it does not.

Figure 3. The layers compose — possibilism runs first and bounds the Bayesian measure-uncertainty

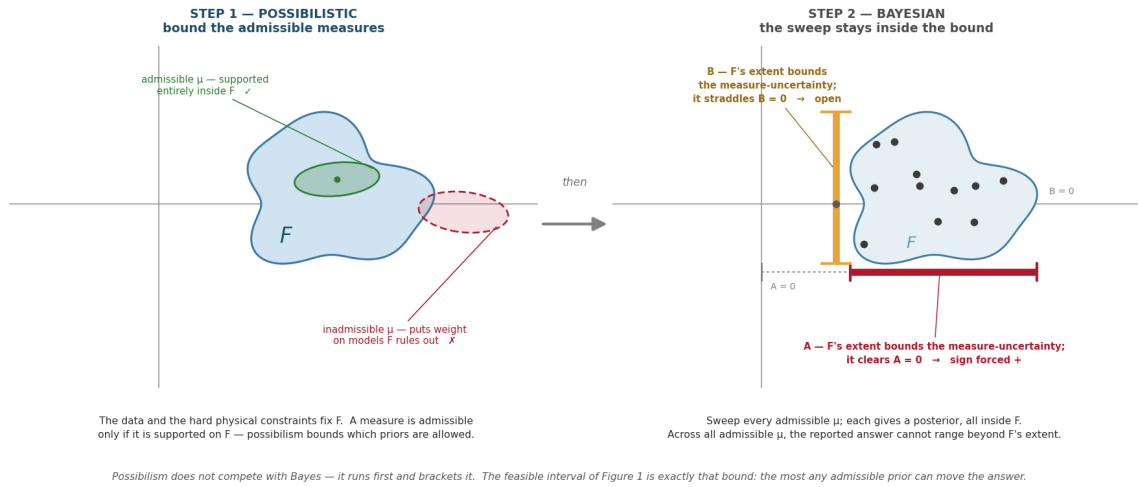


Figure 3: The layers compose. Possibilism runs first and bounds the admissible measures; the Bayesian sweep then stays inside that bound. The feasible interval below is exactly that bound.

interval $[a_{\min}, a_{\max}]$ of the anomaly across the ensemble. That interval is F projected onto the cell's axis—exactly the bound Figure 3 draws:

- $a_{\min} > 0$ everywhere in the ensemble $\rightarrow$ **forced-high** (no data-consistent model makes this cell non-positive);
- $a_{\max} < 0 \rightarrow$ **forced-low**;
- the interval straddles zero $\rightarrow$ **measure-dependent** (the sign itself is not data-forced);
- the interval sits inside a small band around zero $\rightarrow$ **forced-quiet**.

Figure 4 puts the classification on a concrete ensemble. The same ensemble, read two ways: the probabilistic reading collapses it to a mean and an error band; the possibilistic reading keeps the feasible interval and classifies it. The big feature is forced; the flanks and the small bump are measure-dependent—and the probabilistic band does not distinguish them.

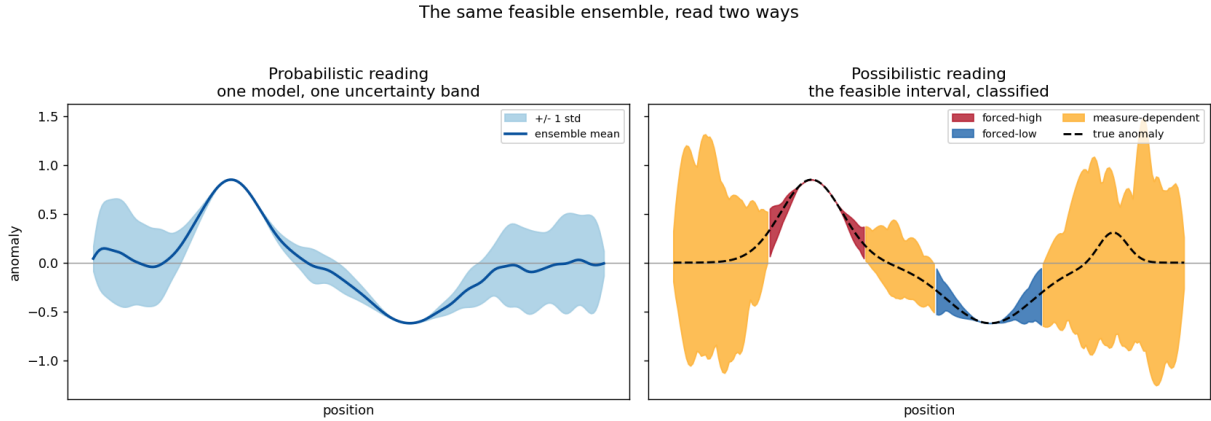


Figure 4: The same feasible ensemble, read two ways. Left: one model, one uncertainty band. Right: the feasible interval, classified into forced-high / forced-low / measure-dependent.

On prior art — said straight. This is not unprecedented, and pretending otherwise would not survive five minutes of your scrutiny. Computing bounds on what a model *can* be, rather than a single estimate, is the spirit of Backus–Gilbert extremal inversion (1968); the under-determined directions are the null space of resolution analysis; multi-model joint coupling has the Gramian-constraint literature (Zhdanov, 2012 onward). And a transdimensional / reversible-jump MCMC posterior ensemble (Bodin & Sambridge, 2009) already *contains* this information—forced structure appears as high-consensus, measure-dependent structure as sign-variable posterior mass. The epistemic intuition is old. What I am putting forward is narrower: not a new inverse-theory principle but a *reporting discipline*—the forced / measure-dependent split made an explicit, first-class output; measure-dependence treated as the operational diagnostic of *underdetermination*; and the whole run as a transferable, documented procedure on top of arbitrary inversion machinery, rather than the informal robustness check practitioners already perform inconsistently. The contribution is the operationalization, not the distinction.

3 Method

The feasible set. A model is *feasible* if it (a) fits the data to within the noise level and (b) satisfies the hard physical bounds. The bounds are not left implicit: `geophysical_invariants.md` stratifies them into Tier-1 (frame-independent—positivity, first-arrival minimality, a generous velocity envelope; used freely), Tier-2 (frame-dependent—typical mantle/crust values, reference Earth models, petrophysical relations; soft priors only), and Tier-3 (the observational context). The possibilistic feasible set is bounded by Tier-1 alone. Worth noting in passing: TFM currently clamps node velocities to 7.5–9.5 km/s and initializes from IASP91—Tier-2 values used as hard limits. That is the layer-confusion the stratification is meant to catch.

Sampling it. Generate many feasible models by inverting toward many random reference models, each inversion taken to the point where its misfit equals the noise level. The references diversify the data-null directions; the data-resolved directions are common to all. Then read the feasible interval and classify it (§2).

The decomposition is forward-model-agnostic. It operates on the ensemble of models; it does not care how they were produced. That is the point of the two demonstrations below: the *same* decomposition code runs on a linear straight-ray operator and a nonlinear Eikonal operator.

4 Demonstration 1 — straight-ray operator (linear)

A synthetic velocity model: one large tilted high-velocity slab, one broad low-velocity zone, three small sub-resolution blobs. Synthetic first-arrival times with 1.2% noise, dense four-edge ray coverage. The feasible set is sampled exactly per §3 (the linear operator admits an exact eigen-decomposition, so the feasible set is parametrized cleanly).

Figure 5 is the result. The forced-sign cores sit on the true features—panel (e), the black forced-high contour lies inside the true slab, the white forced-low contour inside the true low-velocity zone. The numbers:

- **Forced cores are correct.** ~ 7 sign errors within the resolution length out of ~ 300 forced cells ($\sim 2\%$). What the method certifies, the ground truth bears out.
- **The measure-dependent shell captures $\sim 89\%$ of the sub-resolution blob cells**—it correctly flags the detail the data cannot pin down.
- **The forced set is conservative:** $\sim 19\%$ of the grid is sign-certain. This is not a weakness. It is the method saying out loud what is true—most of a tomographic image is *not* forced.

5 Demonstration 2 — Eikonal operator (nonlinear)

The straight-ray operator is exact only in a homogeneous medium. The faithful operator is the Eikonal first-arrival solver—the operator class your `FMM.cpp` implements. First-arrival rays bend: toward fast structure, away from slow (Figure 6). `eikonal.py` provides it (Fast Marching Method solver + ray-path Fréchet kernel), standalone and self-tested.

Because travel time is now a nonlinear functional of slowness, the one-shot linear solve is replaced by an iterative Levenberg–Marquardt inversion that recomputes the ray paths through the cur-

Possibilistic decomposition of a tomographic inversion ensemble - synthetic demonstration

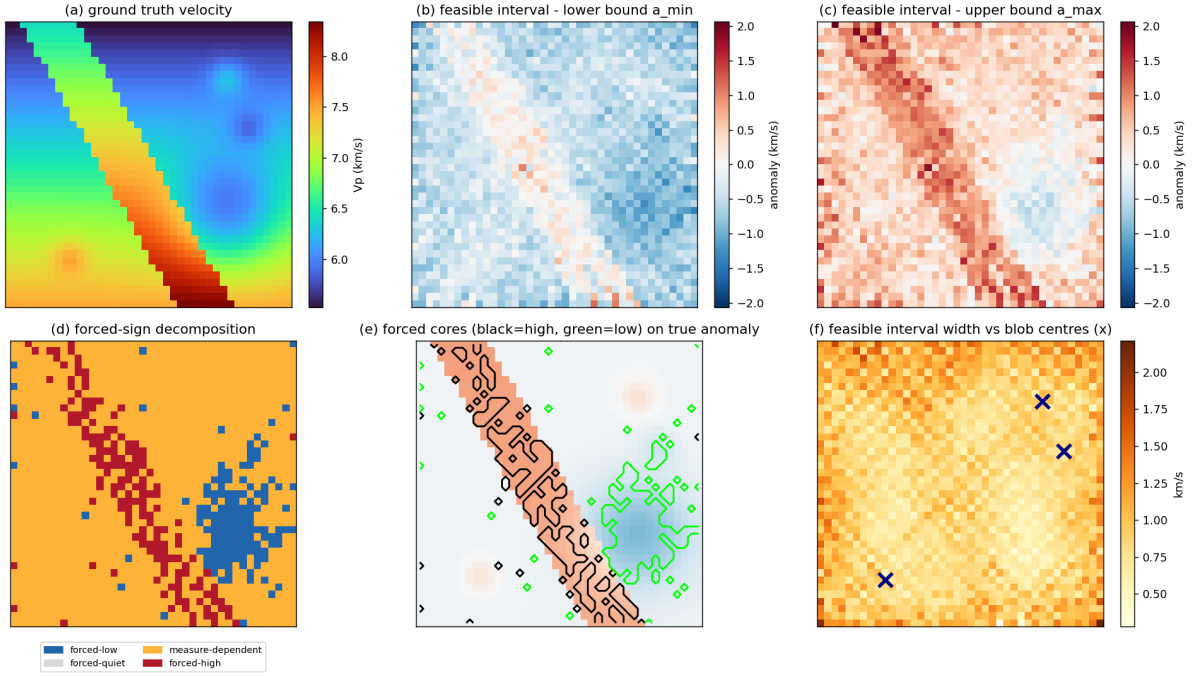


Figure 5: The straight-ray (linear) demonstration. Panel (d) is the decomposition; panel (e) shows the forced cores landing on the true features.

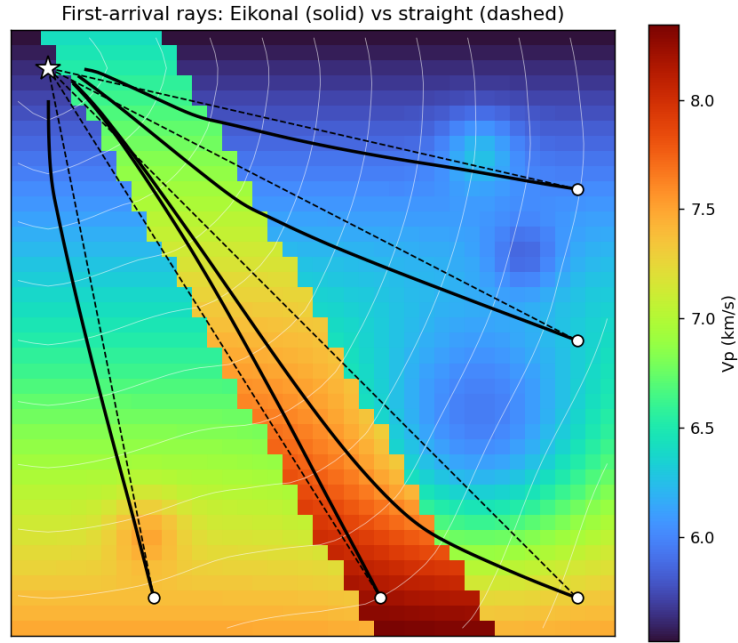


Figure 6: Why the forward operator matters. Solid: Eikonal first-arrival rays, bending through the medium. Dashed: the straight-ray approximation.

rent model—the DGN + FMM structure of your own pipeline. Figure 7 is the result, and the decomposition code is *identical* to Demonstration 1:

- **Forced cores $\sim 89\%$ sign-correct within the resolution length**, but $\sim 71\text{--}81\%$ strict (cell-exact). The gap is resolution-length blur—a forced cell one or two cells off a true feature edge, not a sign mistake—but the strict figure is the honest co-headline; the within-resolution number alone overstates the precision.
- **The measure-dependent shell captures $\sim 76\%$ of the blobs.**

These figures are from the feasible-set sampler at its default operating point, and at that operating point the forced set itself over-claims: an adversarial stress test (§8, point 6; `witness_pass.md`, RWC-2) finds $\sim 41\%$ of forced cells admit a feasible, equally-smooth, opposite-sign model. The decomposition layer is sound and forward-model-agnostic; what is coverage-limited is the feasible-set *sampling* that feeds it (§9).

The decomposition transferred across two genuinely different forward operators without change. That is the load-bearing result of this note: the possibilistic reading is a property of *inversions*, not of a particular operator.

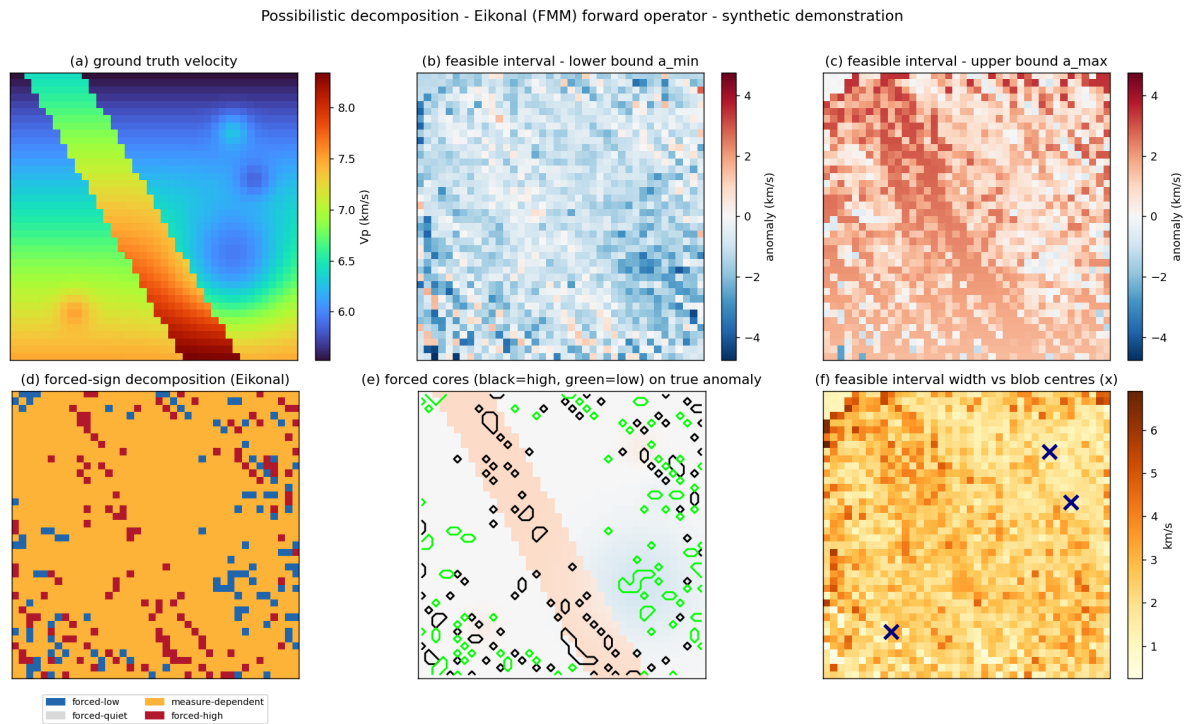


Figure 7: The Eikonal (nonlinear) demonstration—same decomposition, faithful operator.

6 Demonstration 3 — real-data on the Volve walkaway VSP

Demonstrations 1 and 2 validate the methodology against synthetic ground truth. This section asks the next question: does any of it survive contact with a real field dataset? The **Equinor Volve open release** (North Sea, 2018), made available under Equinor’s *HRS Terms and condi-*

tions for licence to data — *Volve* (§6.6), includes walkaway-VSP surveys for two wells with full wireline-log suites — independent sonic ground truth at known locations. The data pipeline from raw SEG-Y to forced/measure- dependent classification is shipped in the *volve/* subpackage of the methodology repository; the report below is what came out of it.

6.1 1D $V_p(z)$ on well 15/9-F-15A — the methodology calibrates

Picks: STA(10ms)/LTA(100ms) trigger with AIC refinement on the Z-component geophone traces, 1215 ok of 1248 (97.4% yield; the 33 no-trigger traces are the 5 far-offset sources). Ensemble: 30 random-reference Eikonal Gauss–Newton members on 45 depth bins (70 m), envelope [1.5, 5.5] km/s, smoothness correlation 250 m. Validation: the LAS bundle’s DT-EDIT sonic curve, checkshot-stretched to TVD.

Headline numbers, real data:

- **Per-bin ensemble V_p interval covers the wireline sonic in 36/45 bins (80.0 %).**
- **Held-out arrival calibration: 228/243 (93.8 %) of held-out pick times fall inside the ensemble’s predicted-time interval.**
- Signed error (ensemble median minus sonic) median +0.29 km/s; the ensemble runs slightly slow at depth, consistent with the first-order FMM solver and the 1D-along-bore parameterization.

That is the “methodology mechanically works on real first arrivals” outcome. Calibration matches the methodology’s claim; the operator-relative caveats of §10 apply unchanged.

6.2 2D joint inversion on F-15A + F-11 T2 — structural misspecification surfaces

Two wells are 1061 m apart laterally; combined they sample depths ≈ 130 –4530 m. Promoting to 2D $V_p(w, z)$ on a (well-line, depth) grid through both wellheads multiplies the model space $\sim 180 \times$ (45 bins $\rightarrow$ 8004 cells) without $180 \times$ more data, and changes the inversion’s character.

Three rounds of work were needed to get an honest read:

1. **First-cut report.** A 12-member ensemble $\times$ 2 Gauss–Newton iterations produced sonic-inside coverage of 10% at each well’s wellhead column, a joint held-out inside rate of 23%, and 79% “forced-quiet” cells. The narrative on first read was “*prior dominance in unilluminated cells.*”
2. **External review.** A scoped brief was circulated for independent review, which converged on four interpretation artifacts the first read was leaning on: the anomaly baseline was the ensemble mean (gauge-sensitive), the sonic was sampled at the wellhead column instead of along the deviated bore (category error), the ensemble size was small (under-exploration risk), and the prior strength was not directly tested. The review specifically named the structural-misspecification hypothesis — *2D $V_p(w, z)$ on a 3D Earth* — as the leading alternative.
3. **Phase A diagnostics + Phase B re-run.** Each artifact was tested in sequence. Switching to a wireline-derived anomaly baseline collapsed the 79% forced-quiet to 22% (the original number was gauge-sensitive). Sampling the sonic along the bore trajectory raised the F-

Volvo 15/9-F-15A - phase 4 EIKONAL real-data decomposition

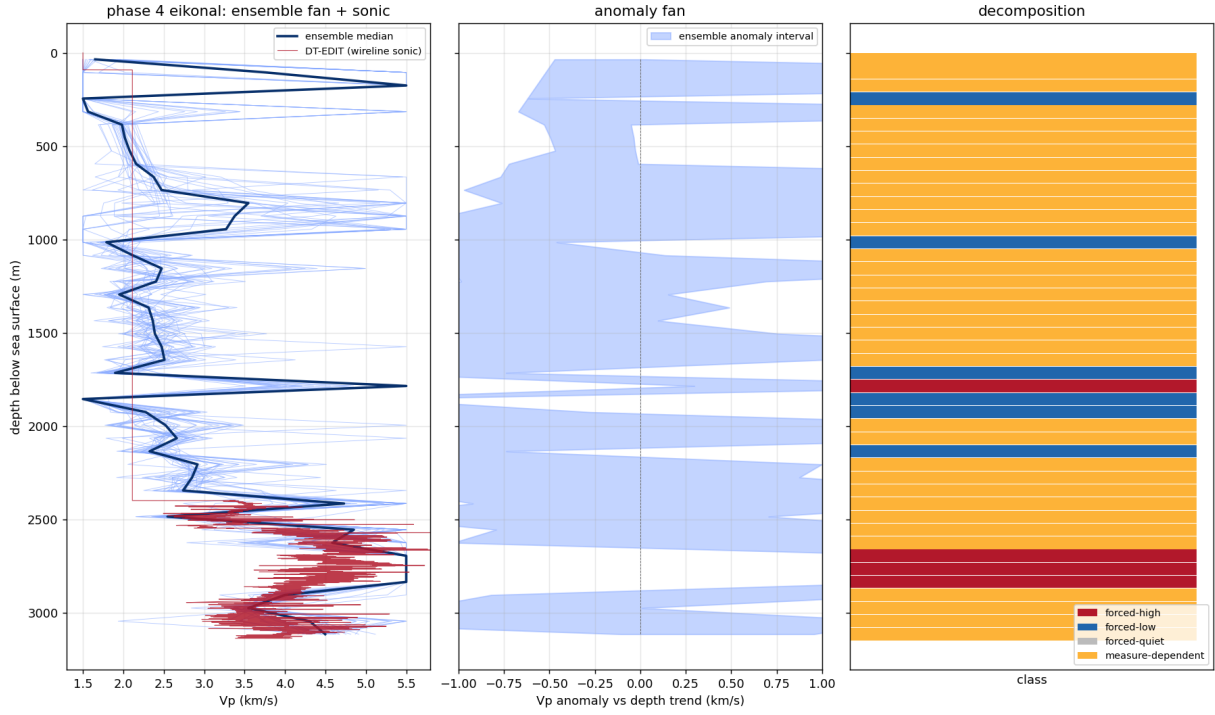


Figure 8: Possibilistic decomposition on the 15/9-F-15A walkaway VSP. Left: 30-member ensemble $V_p(z)$ fan against the DT-EDIT wireline sonic. Centre: the per-bin anomaly interval relative to the depth trend. Right: the forced-high / forced-low / measure-dependent classification. The ensemble's per-bin V_p interval covers the wireline sonic in 80% of depth bins, and the held-out arrival calibration lands at 93.8% — at the rate the possibilistic frame claims for full-coverage forecasts.

15A sonic-inside from 10 % $\rightarrow$ 32 % (post-processing) $\rightarrow$ **57 %** (Phase B re-inversion, 30 members $\times$ 3 GN iters, looser prior). Loosening the smoothness correlation 350 $\rightarrow$ 700 m and the envelope 1.5–5.5 $\rightarrow$ 1.3–6.0 km/s did **not** widen the predicted-time interval (21 ms $\rightarrow$ 24 ms) and did **not** shrink the held-out residual (137 ms $\rightarrow$ 134 ms RMS). Across all 30 members, **44 % of cells classify as forced-low against the wireline-sonic baseline** — the ensemble systematically sits below the sonic, and the diagnosis is stable.

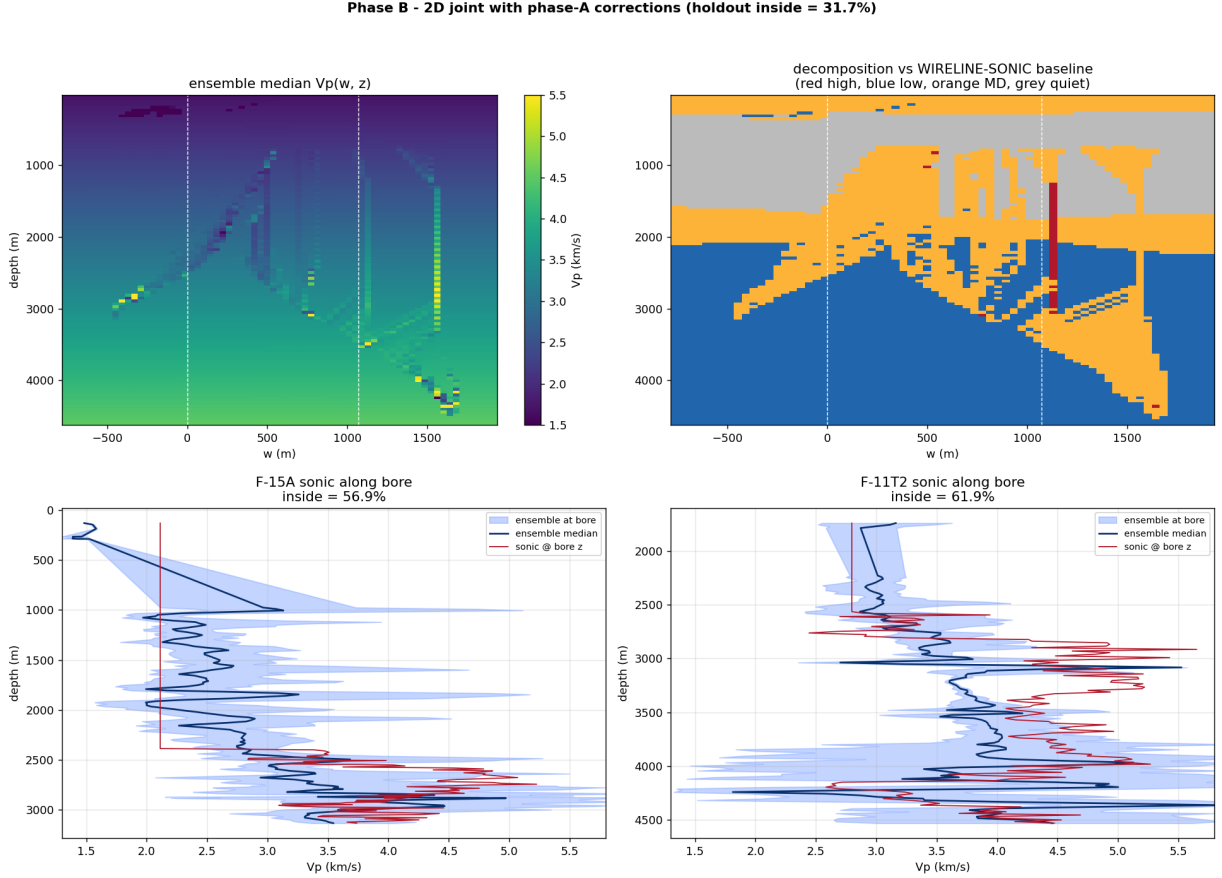


Figure 9: 2D $V_p(w, z)$ joint inversion of the F-15A + F-11 T2 picks (Phase B, with the external-review artifact corrections applied). Top: ensemble median $V_p(w, z)$ and the forced/measure-dependent classification versus the wireline-sonic baseline (red high, blue low, orange measure-dependent, grey forced-quiet). Bottom: sonic-along-bore vs ensemble at each well — 57 % (F-15A) and 62 % (F-11 T2) inside the ensemble interval. The persistent 44 % forced-low (blue) at depth is the diagnostic: 2D $V_p(w, z)$ is structurally insufficient for the 3D Earth at this geometry, and the methodology reports it.

The methodology is doing what it should. The ensemble has ten effective dimensions (it is exploring genuine diversity, not collapsing onto a single member), and yet every member fits the picks but sits below the sonic at depth. The 44 % forced-low survives prior loosening because the binding constraint is the parameterization itself, not the prior strength. *That* is what “the forward operator (or parameterization) is the load-bearing approximation” looks like in real data, and the decomposition correctly flags it.

6.3 External review as part of the method

The review round and the four corrections it produced are themselves part of the methodology contribution. The discipline is: *publish the first-cut interpretation, circulate a scoped brief for independent review, run the diagnostics it recommends, retract the parts that were gauge-sensitive or confounded, re-run with the corrections, and re-report*. This was the second time in the methodology’s development that external review caught an interpretation drift; the practice is now treated as a step in the workflow, not an extra.

6.4 Head-to-head: possibilistic vs Bayesian vs neural-net

On the F-15A 1D problem, under matched physics (Eikonal FMM forward), matched prior (smooth random $V_p(z)$ on 45 bins, envelope 1.5–5.5 km/s, smoothness correlation 250 m), and matched data (the same 1215 picks), three different representations of inversion uncertainty:

Method	Sonic-inside	Holdout median RMS
posdec (feasibility interval, 30-member)	80.0 %	179 ms
MCMC (emcee linearized, 90 % credible)	55.6 %	179 ms
MCMC (95 % credible)	64.4 %	(same)
NN (MLP, MC dropout, 90 % band)	6.7 %	490 ms
NN (MC dropout, 95 % band)	8.9 %	(same)

The point is not “which method is best.” It is that **each choice of uncertainty representation gives a quantifiably different account** of the same inverse problem — same physics, same prior, same data — and the choice is methodologically load-bearing. The Tier-1 sensitivity sweep on the synthetic side flagged the analogous load-bearing-ness of the smoothness prior (§8 item 7, Figure 14); the three-way head-to-head is its real-data analog. Possibilistic is the most conservative; that is the methodology’s design, and on this real-data test it delivers the calibration its frame claims.

6.5 Honest scope of the real-data section

- One field (Volve), two wells. The 1D F-15A result calibrates; the 2D joint result correctly diagnoses parameterization limits.
- Eikonal first-order FMM, not finite-frequency or full-waveform.
- The MCMC comparator uses a linearized forward (matrix–vector per step) at the ensemble-median reference, not a full nonlinear chain; emcee is run with 100 walkers $\times$ 3500 steps.
- The NN comparator is a stock MLP trained on synthetic eikonal-forward picks from the matched prior, MC dropout for uncertainty. NN tomography state-of-the-art (normalizing-flow posteriors, Bayesian NNs, PINNs) was not invoked; that is a separate scope.
- A structurally different parameterization (anisotropy, finite-frequency) was not tested; the full-3D case is taken up in §7. The remaining extensions are best paired with a domain collaborator.

6.6 Data attribution and licence

The data used in this section is the **Equinor Volve open dataset** — the walkaway-VSP bundles for wells **15/9-F-15 A** and **15/9-F-11 T2** under 08.VSP_VELOCITY, and the corresponding

F-15A three-way Vp(z): posdec vs MCMC vs NN under matched physics + matched prior

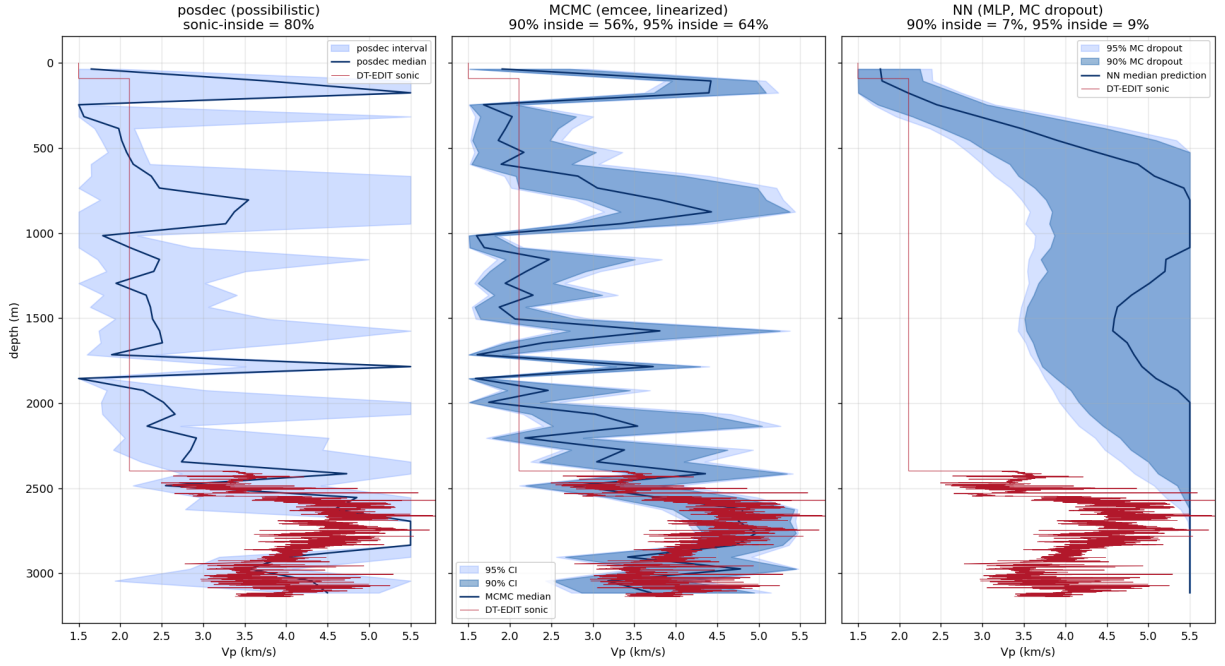


Figure 10: Three uncertainty representations on the same inverse problem. posdec feasibility interval is widest and covers the sonic at the rate the possibilistic frame claims (80%). MCMC’s Bayesian credible intervals contract more tightly because the Gaussian likelihood imposes shape constraints beyond mere feasibility; same central estimate (179 ms holdout RMS) but more confident uncertainty. MC dropout on a supervised MLP trained on matched-prior synthetic data captures epistemic uncertainty within the training distribution and is blind to real-data distributional shift: the dropout band is narrow and the median prediction is also off the real trend (490 ms holdout RMS).

petrophysical-interpretation bundles under 05.PETROPHYSICAL INTERPRETATION. The data is licensed by **Equinor ASA**, **ExxonMobil Exploration & Production Norway AS**, and **Bayerngas Norge AS** (the “former Volve license partners”) under the *HRS Terms and conditions for licence to data — Volve* (the “HRS T&C”). The HRS T&C is a CC BY 4.0-derived licence with two material modifications: (a) the Licensed Material may not be sold, and (b) the licence covers all data in the dataset whether or not it is by law covered by copyright.

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Nothing in this section (or elsewhere in this note) **constitutes or implies endorsement** of this work by Equinor or the former Volve licence partners (HRS T&C §4). The licence partners are credited solely as the data providers; the methodology, the inversion results, and the interpretive narrative are the author’s, with the honest-scope caveats of the previous subsection and of §10 applying.

Citation and link. The official HRS Terms and conditions document is available from Equinor’s open-data portal at <https://www.equinor.com/energy/data-sharing>. The dataset itself (“the Volve Data Village”) is described in Exhibit 1 of the HRS T&C; the specific bundles used here are folders 8 and 9 of that exhibit (`Well_logs` and `Well_logs_pr_WELL`). When citing this work, please also cite Equinor’s release per the HRS T&C attribution requirement.

7 Demonstration 4 — real 3D on the PoroTomo nodal array

Demonstration 3 reached real data, but the acquisition was a walkaway VSP: the inversion was 1D along the bore, then 2D in a single well-line plane. The natural next question (§6.5) is whether the method survives a **genuinely 3D acquisition geometry**, where rays cross a volume from sources and receivers spread over a surface. This section runs the full pipeline on the **PoroTomo** experiment (Brady Hot Springs, Nevada; U.S. DOE geothermal program, March 2016), an open dataset whose geometry is areal, not linear. The 3D pipeline ships in the `porotomo/` subpackage; the data is CC BY 4.0 (§7.6).

What PoroTomo adds beyond Volve, and what it gives up: it has $\sim 130\times$ the picks (165,359 published P-wave auto-picks vs Volve’s $\sim 1,200$), an areal source–receiver layout, and — uniquely — *multiple independent published velocity models and a drilling-based 3D geologic model* to test against. What it lacks is co-located sonic logs; ground truth here is other methods’ models and the geology, not a wireline curve.

7.1 The dataset and the inversion

Sources: 196 vibroseis points swept over four acquisition stages. Receivers: 238 live three-component nodes on a $\sim 1.7 \times 1.9$ km grid. The travel-time picks are the project’s own AIC auto-picks (GDR 924), so the inversion input is a 10 MB table, not a terabyte of waveforms. **Stage 1 (41,971 picks after an apparent-velocity outlier filter) is inverted; stage 2 is held out untouched for calibration; stages 3–4 are reserved for the time-lapse probe (§7.5).** Forward operator: the 3D first-arrival Fast Marching solver of `porotomo/c/eikonal3d.c` (a dependency-free C kernel, the 3D sibling of `eikonal.py`), with analytic source-ball seeding to suppress the first-order scheme’s near-source error. Ensemble: 30 random-reference damped Gauss–Newton members on a 50 m grid ($13 \times 50 \times 47$; topography masks air cells out of the model), each fit to a 60 ms noise floor — the achievable floor set by the auto-picks’ robust repeat scatter (~ 36 ms) plus their heavy outlier tail (p95 $|\Delta t| = 140$ ms) and the first-order forward error. All 30 members converge to 53–63 ms RMS.

One negative result drove a method fix and is worth stating. The first full ensemble damped only toward the random reference, with no explicit roughness penalty — and produced *inverted* interval-width ordering: illuminated cells came out **wider** (4.8 km/s median) than unilluminated ones (1.8), because members streaked bound-to-bound along the rays. In 3D the cells-per-datum ratio is high enough that the implicit smoothness of the reference is not enough; the smoothness class has to be made explicit. Adding a graph-Laplacian roughness penalty on the deviation from the reference (the declared class) fixed the ordering (0.57 lit vs 1.50 unlit), removed all bound-pinning, *improved* the fit, and — because it conditions the linear solve — cut per-member cost $\sim 20\times$. This is exactly the load-bearing-ness of the smoothness prior that the Tier-1 sensitivity work flagged on the synthetic side (§8 item 7), reappearing as a hard failure mode at 3D scale. The fix is honest about what it is: a *declared* class, whose forced labels are then reported relative to it and stress-tested for sensitivity to it (§7.5).

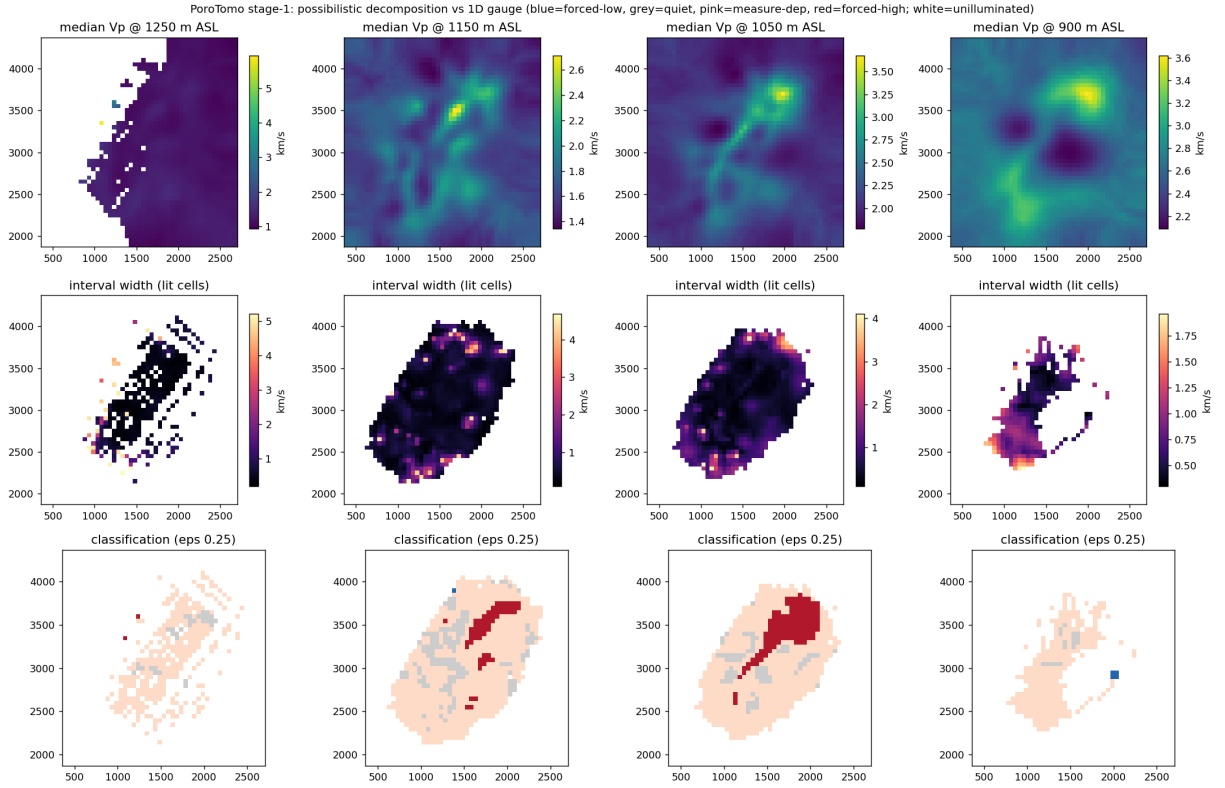
7.2 The decomposition

The gauge is a laterally-uniform 1D median profile — a physical null (“no lateral structure”), so the forced-high/low labels are claims about *lateral* anomalies the data demands, and are non-tautological (unlike an ensemble-mean background — the gauge error the Volve phase-5 external review caught and §6.3 documents). Cells without ray coverage are flagged unilluminated and held out of the forced statistics rather than allowed to pose as findings; on this geometry 20.7% of ground cells are illuminated.

At the deadband $\varepsilon = 0.25$ km/s, over the illuminated volume: **7.7% forced-high, 0.1% forced-low, 13.1% forced-quiet, 79.1% measure-dependent** (the full ε sweep is in `porotomo_decomposition_cert.json`). The forced-high cells are not scattered — they form a contiguous NE-trending fast body at 50–150 m depth (Figure 11). The honest headline is the 79%: most of what a single deterministic model would render as confident structure is, under this data and prior class, *permitted but not forced*. The method’s job is to say which 8% is not.

7.3 Calibration: held-out arrivals and independent models

Two tests, neither using a sonic log. **Stage-2 holdout:** the ensemble-median predicts the untouched stage-2 arrival times at 53.9 ms RMS, versus 53.1 ms in-sample — essentially zero generalization gap, the direct evidence that the ensemble has not overfit. (Raw inside-interval is 27%,



rising to 65 % with a ± 36 ms pick-noise allowance; unlike Volve, the 3D predictive intervals are *tight relative to pick noise* — median 38 ms — so the raw rate here mostly measures the auto-picks’ own noise, and the in-sample $\approx$ holdout RMS equality is the load-bearing number.)

Independent published models (Figure 12). The PoroTomo team’s final property grid (GDR 1124) carries six V_p columns from three groups and two method families — five distinct models, since the two later Thurber entries are byte-identical in the published grid. Against the illuminated cells, the feasible set brackets them: the Thurber and Nayak travel-time/body-wave models fall inside our per-cell intervals in **95–98 %** of cells within 0.25 km/s, with correlation $r = 0.89$ –0.92; the closest model (Thurber 2017-11) matches our field-wide median to 0.004 km/s and the family sits within 0.07. Where we recover the same earth, we recover it almost exactly. The two systematic departures are interpretable, not failures: at the slow extremes our declared smoothness class compresses contrast the finer published parameterization resolves, and the Matzel *sweep-interferometry* model runs ~ 0.3 km/s fast throughout — different physics sampling different structure, divergent from the travel-time models too.

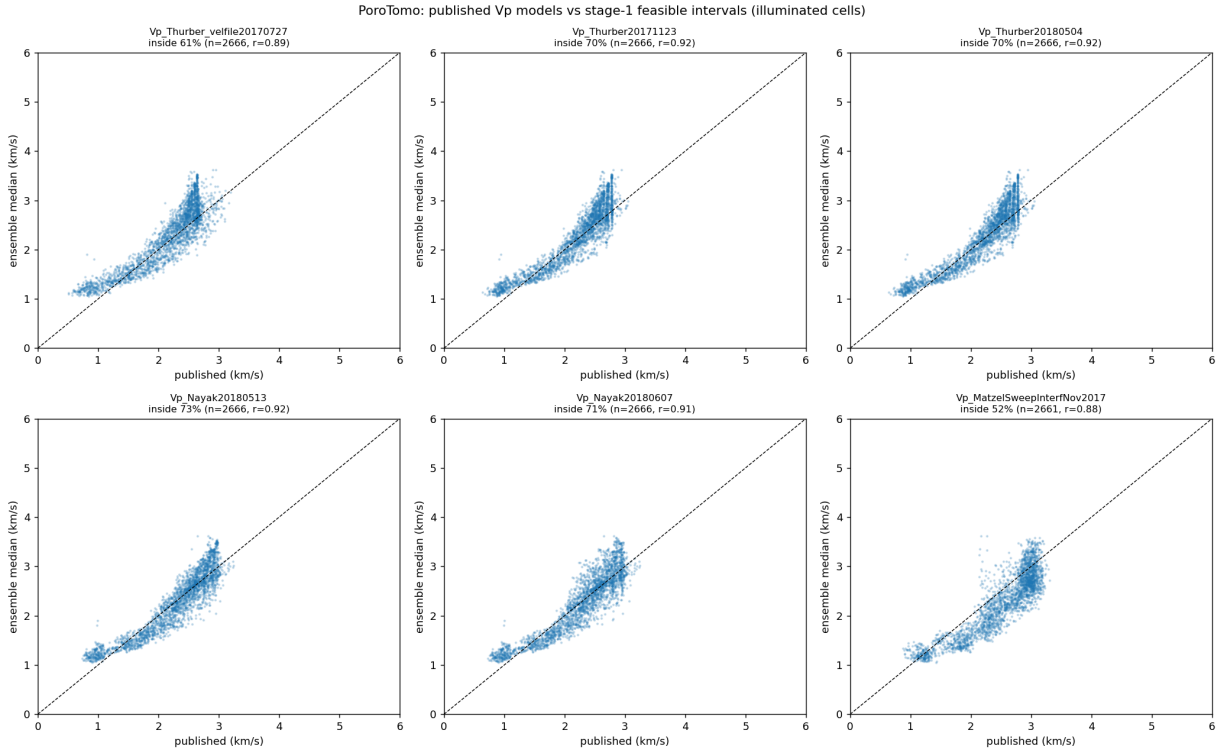


Figure 12: The six published PoroTomo V_p columns (five travel-time/body-wave panels — of which the two later Thurber entries are byte-identical in the published grid — plus one sweep-interferometry model) vs the stage-1 feasible-interval ensemble median on illuminated cells; inside-interval fraction and correlation annotated per panel. Travel-time and body-wave models sit inside the intervals at 95–98 % (± 0.25 km/s) and track the ensemble median tightly; the interferometry model’s systematic offset is a real, localizable physics difference.

7.4 Three-way, under matched everything

The §6.4 head-to-head repeats in 3D, now against a Bayesian comparator built for scale: with 27,234 unknowns, an affine-invariant MCMC sampler is infeasible, so the Bayesian baseline is **randomize-then-optimize (RTO)**, which draws exact samples from the linearized-Gaussian posterior — matched physics, noise, and prior *class* to the possibilistic ensemble. The neural baseline is the same stock MLP-with-MC-dropout recipe as Volve, trained on synthetic eikonal forwards from the matched prior. All three face the identical stage-2 holdout and published-model tests (Figure 13):

Method	Thurber-in (± 0.25)	Holdout-in (± 36 ms)	Median width	Holdout RMS
posdec (feasible interval, 30)	98 %	65 %	0.57 km/s	54.0 ms
Bayes (RTO, 95 % credible)	90 %	52 %	0.26 km/s	54.0 ms
NN (MLP, MC-dropout, 95 %)	90 %	58 %	0.69 km/s	61.9 ms

The ordering matches Volve exactly. posdec and the Bayesian posterior share physics and so share the central estimate (identical 54.0 ms RMS), but the 95 % credible intervals are *half the width* and miss accordingly — textbook credible-interval overconfidence once the Gaussian likelihood imposes shape beyond mere feasibility. The NN’s dropout band is *wider* than the feasible interval yet covers less on every test — a miscalibrated shape, not just a miscalibrated scale. Possibilistic is again the representation that delivers the coverage its frame claims, on a real 3D problem, against independently-built models.

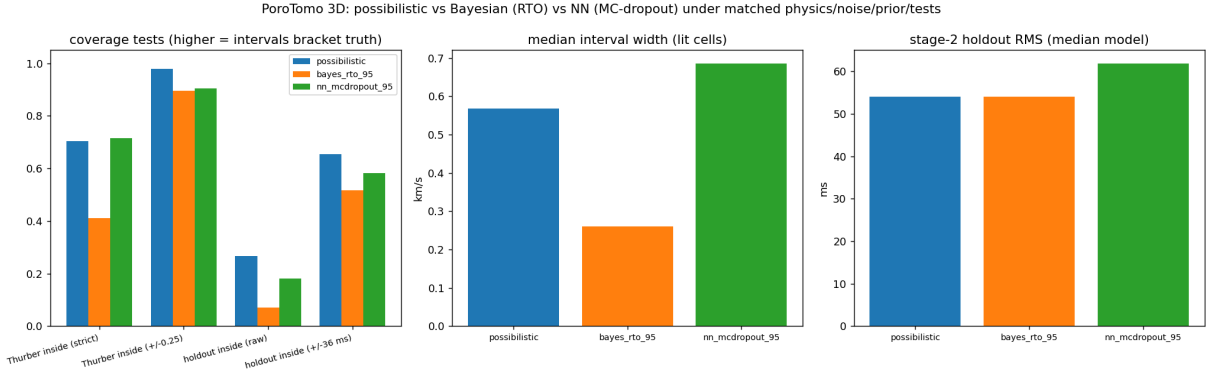


Figure 13: PoroTomo 3D, three uncertainty representations under matched physics/noise/prior/tests: coverage tests (published-model-inside and stage-2 holdout, higher is better), median interval width, and stage-2 holdout RMS. posdec leads every coverage test; the Bayesian posterior matches its central estimate but is overconfident; the NN is wide and miscalibrated.

7.5 Three stress tests: sensitivity, time-lapse, geology

Three further probes, run before this section was written, are reported in full in the repository’s docs/ companion document; the load-bearing results:

- **Smoothness-class sensitivity** (the Tier-1 protocol of §8 item 7, in 3D). Re-running the ensemble at perturbed class hyperparameters and measuring the Jaccard overlap of the forced sets against a half-ensemble control ceiling ($J = 0.70$): the forced-high set is robust

to *doubling* the correlation length ($J = 0.67 \approx \text{ceiling}$) but moves under *halving* it ($J = 0.46$) and under changes to the smoothness *ratio* ($J = 0.30\text{--}0.36$). The forced claims are class-relative — which is precisely how the method states them, and why this probe is part of it, not an afterthought.

- **Possibilistic time-lapse** — a question type no treatment of this dataset has posed that we could find in the published record. Inverting stages 2/3/4 with *paired references* (shared member seeds, so the prior spread cancels in the per-cell difference) and using the minimal-change stage 1→2 pair as a null control (a symmetric 2.4 % forced-change false-positive floor): the forced-*slower* rate grows monotonically across stages (1.24 % → 1.88 % → 2.45 %, i.e. 2.0× the null forced-slower rate by stage 4) while forced-*faster* stays at the floor. The honest scope is that per-cell changes are mostly *open* — the expected pumping-induced velocity change ($\lesssim 0.07$ km/s) sits below the per-cell feasible sensitivity (~ 0.45 km/s) — but the asymmetric drift is a real, directionally-consistent forced signal.
- **Geologic cross-tabulation.** Against the Siler & Faulds 3D geologic model (USGS SIM 3469; built from 24 wells and mapping, independent of the seismic data), the forced-high cells are over-represented in mapped colluvium (1.6×) and playa-fringe deposits (1.9×) and nearly absent from old-alluvium and diatomite-fan units (0.03–0.19×) — i.e. the forced set has non-uniform lithologic composition rather than being lithologically flat. This is consistent with the forced label carrying geologic information, but it is not clean corroboration: the forced-high body and the illuminated cells are both shallow-biased, so depth-correlated units could be partly a coverage confound. A confound-controlled version (matching on depth/illumination) is left as future work.

7.6 Honest scope and data attribution

Scope. First-order FMM, not finite-frequency or full-waveform. The picks are the project’s AIC auto-picks, seeded from windows around a preliminary inversion — so a picker-systematic error would be common-mode between the inversion and the stage-2 holdout, and that test cannot catch it (the same status the Volve picker had on both sides). The smoothness class is declared, and its forced labels are reported as class-relative with the §7.5 sensitivity attached. Ground truth is other methods’ models and the geology; there is no co-located sonic log at Brady. The Bayesian comparator is RTO on the linearized forward, not a full nonlinear chain; the NN is a stock recipe, not NN-tomography state of the art.

Attribution. The data is the **PoroTomo** dataset (Poroelastic Tomography at Brady Hot Springs), collected under U.S. Department of Energy Office of Energy Efficiency and Renewable Energy funding and released through the DOE **Geothermal Data Repository** / Open Energy Data Initiative under **Creative Commons Attribution 4.0**. The specific products used: nodal-array P-wave auto-picks and array metadata, GDR submissions 924 (DOI 10.15121/1787666) and 826; vibroseis source timing, GDR 824; the final material-property grid with its six V_p columns, GDR 1124 (DOI 10.15121/1501544); and well lithologies, GDR 490. Per CC BY 4.0, this work is an independent adaptation; it is not endorsed by the PoroTomo team, the GDR, or the DOE, and any errors are the author’s. The deterministic travel-time tomography baseline is Parker et al. (2018, *SRL* 89(5), DOI 10.1785/0220180085); the geologic model is Siler & Faulds (USGS SIM 3469).

8 What building it surfaced — the discipline the method needs

A method that always says “yes” is a gimmick. This one has real failure modes, and finding them is how I know it has content. Each was surfaced by a synthetic run failing honestly; each is now documented in the script headers.

1. **The ensemble must sample the feasible set’s diversity, or it certifies its own bias.** An ensemble built by varying damping *strength* alone shares a bias in the data-null directions; the intersection then stamps that shared bias “forced.” The fix is many random references.
2. **Sample where the data is actually fit**—at the misfit = noise level. An over-damped ensemble has had data-resolved structure regularized away.
3. **“Forced” is a sign statement off the interval, not a hard threshold.** A single anomaly-magnitude cutoff is brittle; the feasible interval is the honest object.
4. **Precision is honest only at the resolution length.** Every feasible model inherits the same forward-operator blur; the forced core is correct to within roughly one cell, not cell-exact.
5. **Feasible models must be physically plausible — smooth.** The raw null space of a ray (line-integral) operator is dominated by high-frequency checkerboard modes the operator cannot see. Those fit the data but are not admissible Earth models; perturbing freely in them injects unphysical speckle. A feasible model is data-fitting *and* smooth.
6. **The forced set is coverage-gated — and the gate is measurable.** A forced label asserts that *every* feasible model agrees; an ensemble that under-samples the feasible set will agree spuriously and over-claim. Two checks quantify this (`witness_pass.md`, RWC-1 and RWC-2): the forced set converges only above an ensemble-coverage threshold—here $N \approx 250$ sampled models for a 40×40 grid—and at the default operating point an adversarial stress test finds $\sim 41\%$ of forced cells *false-forced* (a feasible, equally-smooth, opposite-sign model exists). A forced claim is honest only with its coverage-adequacy curve attached.
7. **The Tier-1 admissibility bounds are themselves load-bearing — and the dependence is now measured.** The possibilistic layer is defined relative to a hard admissibility envelope (`geophysical_invariants.md`): a velocity range, a positivity condition, a smoothness preference enforced by the sampler. The split is only as data-forced as those bounds are data-independent. The natural worry that the smoothness preference sneaks regularization into the possibilistic layer is now answered quantitatively (`sensitivity_tier1.py`, Figure 14). Tightening the velocity floor $V_{P,\min}$ from 2.0 to 3.0 km/s drops the forced-high Jaccard to 0.27; the $V_{P,\max} = 9$ km/s ceiling is saturated (every member touches it); and the smoothness preference is load-bearing — at the 25% smoothness percentile the smoothness-kept ensemble gives $J(\text{forced-high}) = 0.36$ versus $J = 0.77$ for a random-size-matched control, a ~ 0.4 Jaccard gap that is *specifically* the smoothness preference rather than the generic smaller-sample narrowing. The Tier-1 envelope is therefore reported alongside the coverage-adequacy curve, not assumed.

These are not patches. They are the methodology’s anti-anchoring discipline—the same one that, in the Closure programme, keeps a derivation from quietly tuning itself to the answer it wants.

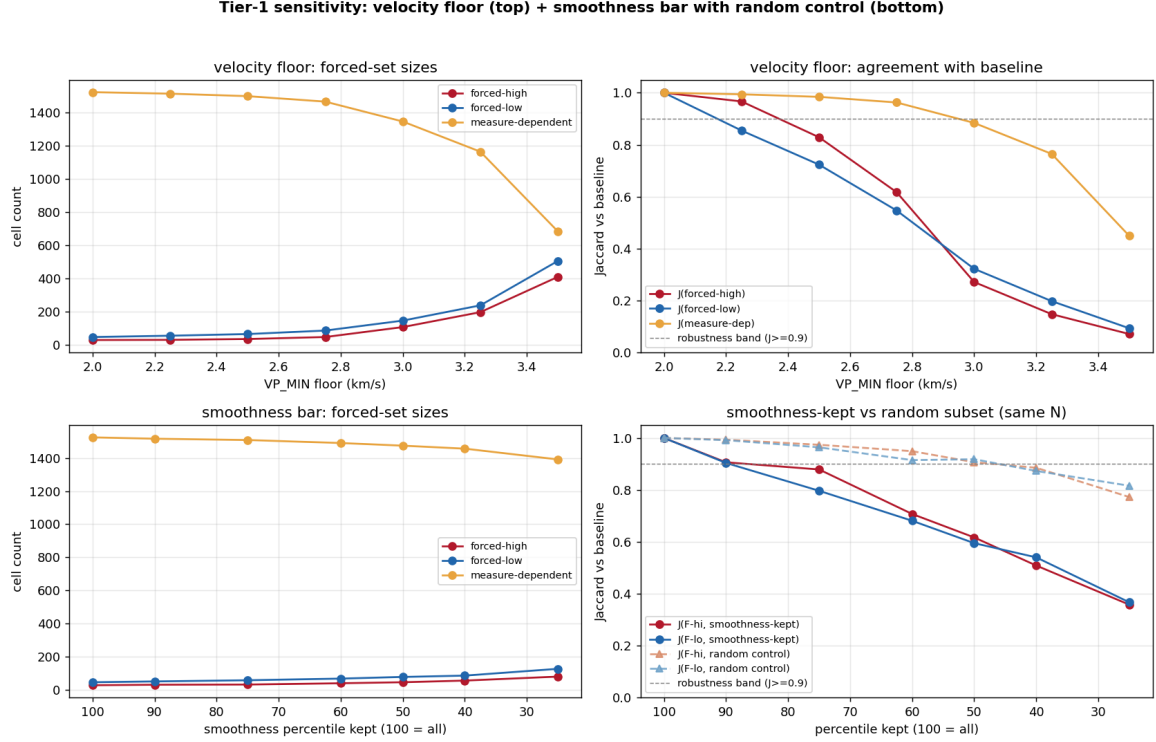


Figure 14: Tier-1 sensitivity sweep on the 396-member RWC-1 ensemble. Top row: velocity-floor sweep (the $V_{P,\max}$ ceiling is saturated at 100% and not separately sweepable). Bottom row: smoothness-percentile sweep with a random-subset control of the same N . The smoothness-kept Jaccard sits below the random-control curve at every p , identifying the smoothness preference as a load-bearing admissibility choice — not merely a sampler convenience.

9 The open frontier — and where you come in

Here is the honest seam, and it is the reason this is a note to *you* and not a finished claim.

The linear straight-ray case has a clean, exact answer: the operator is fixed, $G^\top G$ eigendecomposes once, and the feasible set is parametrized directly. The nonlinear case does not. Sampling the feasible set of a nonlinear inverse problem—getting genuine diversity in the data-null directions without injecting artifacts—is itself a hard problem. Building Demonstration 2 walked straight into it: a naive Levenberg–Marquardt ensemble shares an inversion-dynamics bias; fixing that needs explicit, physically-constrained feasible-set sampling. I have a working version, but I would not call the nonlinear feasible-set sampler solved.

This is exactly the territory of transdimensional / reversible-jump MCMC tomography (Bodin & Sambridge, 2009 onward)—methods built precisely to produce a genuine posterior ensemble. And it is where ZTM already lives: your `runs=20` top-level Monte Carlo *is* a feasible-set sampler. A coverage study on the synthetic problem (`witness_pass.md`, RWC-1) now puts a number on the question: the forced set stabilizes only near $N \approx 250$ sampled models, and a 20-run ensemble sits at roughly three times the converged forced-set size—20 runs is far short of coverage adequacy. So the open question is no longer *whether* the sampler needs upgrading but *to what*: whether a transdimensional sampler reaches the coverage the forced/measure-dependent split requires, on

real data, at tractable cost.

That question is not mine to answer from the outside. It needs your inversion expertise, your code, and your data. My contribution is the possibilistic reading and the formal scaffolding behind it; yours is the geophysics and the instrument. Neither half is sufficient alone. That is the collaboration I am proposing—not “adopt this method,” but “here is a method, validated in the linear case, and here is the concrete nonlinear problem your own pipeline is already engaging. Let us solve that part together.”

10 Honest scope

- The original demonstrations (§§4–5) are **synthetic** — chosen so the method could be validated against known ground truth. §6 promotes the same machinery to **real data** on the Equinor Volve walkaway VSP, with two wells and independent wireline sonics as ground truth; honest scope on §6 is reported inline (§6.5). The 1D F-15A result calibrates at the rate the methodology claims (80 % sonic-inside, 94 % held-out arrival inside), the 2D joint result correctly diagnoses parameterization limits, and the three-way comparison (posdec / MCMC / NN) under matched physics shows that the choice of uncertainty representation is methodologically load-bearing. §7 extends the real-data validation to a genuinely 3D acquisition (PoroTomo), with its own inline scope (§7.6).
- The Eikonal solver is **first-order FMM**. Its $\sim 2\%$ mean accuracy is fine for a forward-model-agnostic demonstration; your FMM-VFD hybrid is the production-grade version.
- The synthetic demonstrations are two-dimensional, with modest velocity contrasts and a single noise realization; §6–§7 are where the machinery is tested on real data at field scale, including full 3D (§7).
- The forced/measure-dependent split is **operator-relative**: it reports what is forced *given a forward operator*. An operator with systematic error propagates that error into the forced set. This is not a defect of the method—it is the method correctly reflecting the operator—but it means the operator’s fidelity is load-bearing.
- The **forced set is coverage-gated**. It is trustworthy only above an ensemble-coverage threshold (RWC-1: $N \approx 250$ here); below it the sampler over-claims (RWC-2: $\sim 41\%$ of forced cells false-forced at the default operating point). The demonstrations are reported at that default point and so over-state the forced set. The decomposition layer is sound—what is coverage-limited is the feasible-set *sampling* that feeds it; a forced claim must travel with its coverage-adequacy curve.
- The forced/measure-dependent split is **conditional on the Tier-1 admissibility bounds**. Both the velocity envelope (the $V_{P,\max} = 9$ km/s ceiling is saturated and the floor is binding well above the methodology’s generous default) and the sampler’s smoothness preference shift the decomposition quantifiably (§8 item 7, Figure 14, `sensitivity_tier1.py`). A published forced/measure-dependent map travels with its Tier-1 envelope *and* its coverage-adequacy curve; either, varied within a plausible range, changes the result.

Acknowledgements

I thank **Brian Crabtree** for helpful conversations. The errors that survived are mine.

Appendix A — The split as an instrument

The main text draws the possibilistic / probabilistic split in model space— F as a set, a feature forced or measure-dependent according to where F 's projection falls (Figures 1–3). Readers who think in terms of estimators and their inputs may find the same split crisper drawn as an *instrument*. The possibilistic report is a function of F alone. The Bayesian report is a function of F and a measure μ —a knob the data never set. Turn the knob and a measure-dependent feature's reported sign swings across zero; a forced feature's reading does not move, because the knob is not wired to it. Nothing here is new—it is Figure 2 again, in a different register—but the register makes one thing plain: the disagreement two defensible analysts can reach is located entirely in the knob, and the knob is not data.

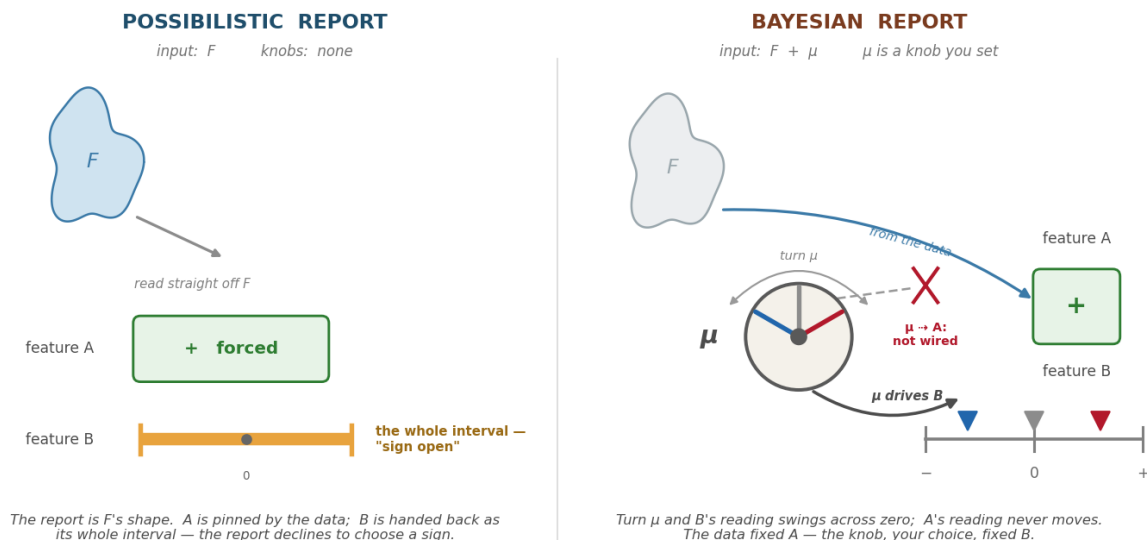


Figure A.1. The report is F , or F plus a knob the data never set.

Figure A.1: The possibilistic / Bayesian split as an instrument. The possibilistic report is a function of F alone—no knob. The Bayesian report needs a measure μ , a knob the data did not set: turning it swings feature B's reported sign across zero, while feature A's reading does not move—the knob is not wired to it.

Pointers

- `geophysical_invariants.md` — the stratified constraint set.
- `synthetic_demo.py`, `synthetic_demo_eikonal.py` — the two demonstrations.
- `eikonal.py` — the standalone Eikonal forward operator (self-test: `uv run python eikonal.py`).
- `decomposition_exact.jl` — the decomposition layer formalized in exact `Rational{BigInt}` arithmetic (Julia); `classify` proven total, exclusive, and exhaustive, the feasible-interval properties exact-verified.

- `c/` — a dependency-free C port of the whole method: the forward operators, the Levenberg–Marquardt inversion over a hand-rolled dense-Cholesky module, the feasible-set sampler, and the decomposition, pulled together by `possibilistic_inversion.c`.
- `volve/` — the §6 real-data pipeline: geometry decoders, picker, per-phase inversions, Phase B re-runs, and the MCMC + NN comparators.
- `porotomo/` — the §7 3D real-data pipeline: pick/station loaders, the 3D FMM C kernel (`c/eikonal3d.c`) with its ctypes wrapper, the 3D ensemble inversion (`inversion3d.py`), decomposition + holdout + published-model checks, the sensitivity/time-lapse/geology probes, and the Bayes-RTO + NN comparators.
- `witness_pass.md` — an adversarial external review of the method, and the two Required Witness Checks it carried, `rw1_forced_stability.py` and `rw2_disconnected_test.py`.
- `inverse_born_methodology.md` (Closure Forces Structure programme) — the source of the two-layer possibilistic / probabilistic discipline.
- Bodin, T. & Sambridge, M. (2009), *Seismic tomography with the reversible jump algorithm*, Geophys. J. Int. — the transdimensional-sampling reference.
- Abatchev, Z. (2019), ZTM/TFM, UCLA — the joint seismic+gravity inverter this note is in conversation with.